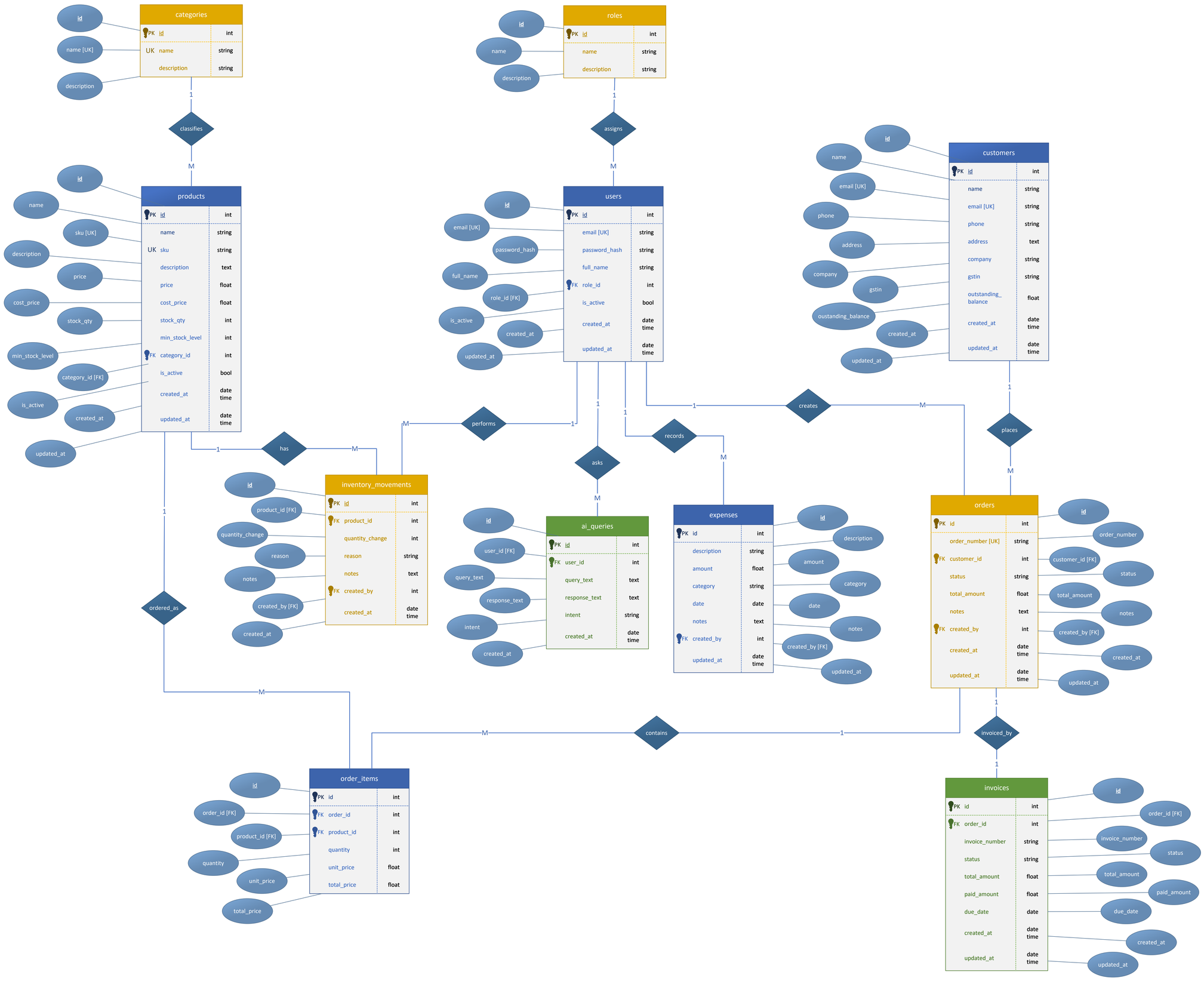
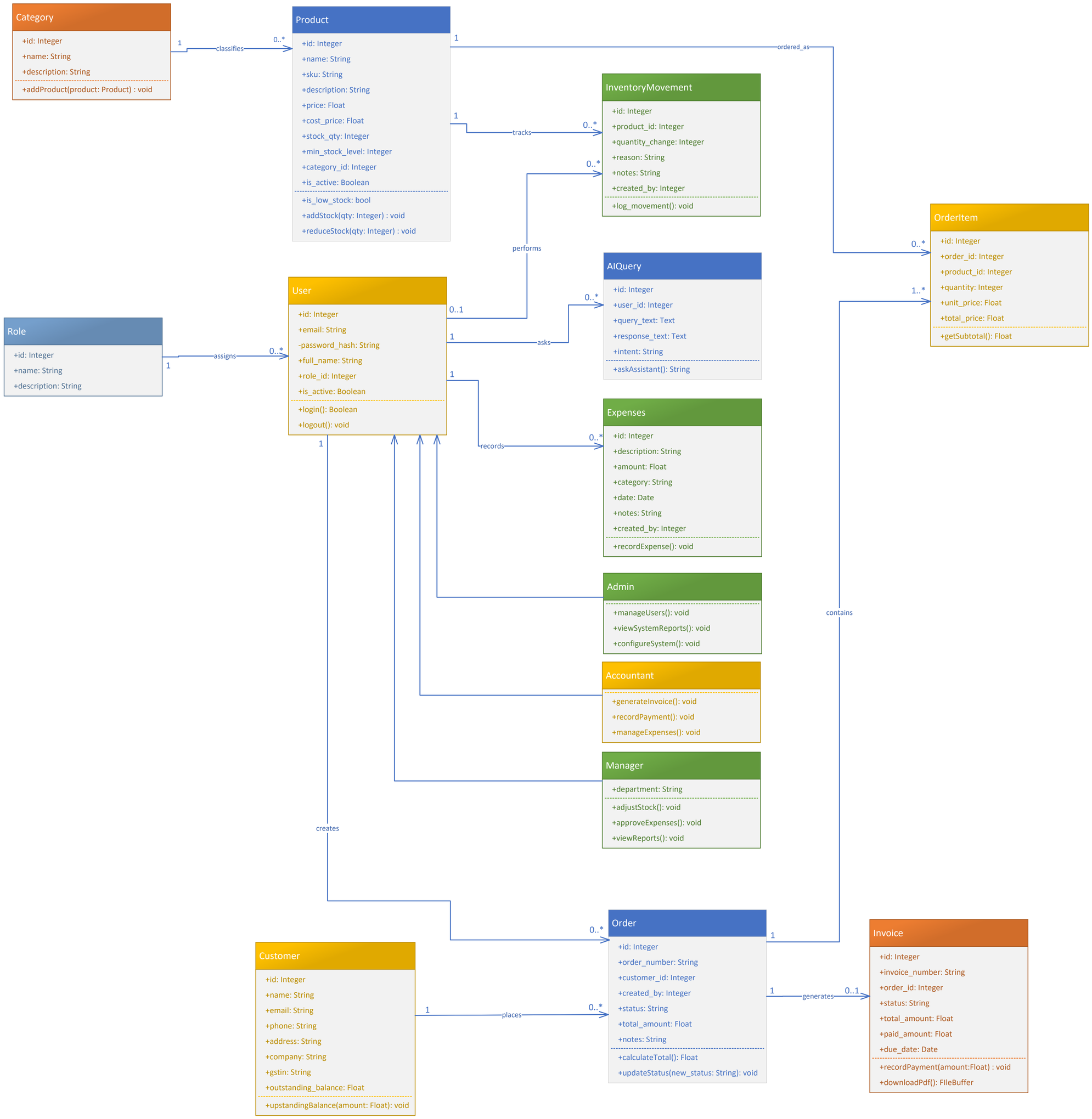

categories
id
PK id int
UK name string
description string
name [UK]
description
1
classifies
M
products
id
PK id int
name string
UK sku string
description text
price float
cost_price float
stock_qty int
min_stock_level int
FK category_id int
is_active bool
created_at date time
updated_at date time
sku [UK]
price
cost_price
stock_qty
min_stock_level
category_id [FK]
is_active
created_at
updated_at
name
description
1
has
M
roles
id
PK id int
name string
description string
name
description
1
assigns
M
users
id
PK id int
email [UK] string
password_hash string
full_name string
FK role_id int
is_active bool
created_at date time
updated_at date time
email [UK]
password_hash
full_name
role_id [FK]
is_active
created_at
updated_at
M
performs
1
1
asks
1
records
M
1
creates
M
customers
id
PK id int
name string
email [UK] string
phone string
address text
company string
gstin string
outstanding_ balance float
created_at date time
updated_at date time
name
email [UK]
phone
address
company
gstin
oustanding_balance
created_at
updated_at
1
places
M
inventory_movements
id
PK id int
FK product_id int
quantity_change int
reason string
notes text
FK created_by int
created_at date time
product_id [FK]
quantity_change
reason
notes
created_by [FK]
created_at
ai_queries
id
PK id int
FK user_id int
query_text text
response_text text
intent string
created_at date time
user_id [FK]
query_text
response_text
intent
created_at
expenses
id
PK id int
description string
amount float
category string
date date
notes text
FK created_by int
updated_at date time
description
amount
category
date
notes
created_by [FK]
updated_at
orders
id
PK id int
order_number [UK] string
FK customer_id int
status string
total_amount float
notes text
FK created_by int
created_at date time
updated_at date time
order_number
customer_id [FK]
status
total_amount
notes
created_by [FK]
created_at
updated_at
ordered_as
1
M
M
contains
1
1
invoiced_by
1
order_items
id
PK id int
FK order_id int
FK product_id int
quantity int
unit_price float
total_price float
order_id [FK]
product_id [FK]
quantity
unit_price
total_price
invoices
id
PK id int
FK order_id int
invoice_number string
status string
total_amount float
paid_amount float
due_date date
created_at date time
updated_at date time
order_id [FK]
invoice_number
status
total_amount
paid_amount
due_date
created_at
updated_at


Category
+id: Integer
+name: String
+description: String
+addProduct(product: Product) : void
1
classifies
0..*
Product
+id: Integer
+name: String
+sku: String
+description: String
+price: Float
+cost_price: Float
+stock_qty: Integer
+min_stock_level: Integer
+category_id: Integer
+is_active: Boolean
+is_low_stock: bool
+addStock(qty: Integer) : void
+reduceStock(qty: Integer) : void
1
ordered_as
1
tracks
0..*
InventoryMovement
+id: Integer
+product_id: Integer
+quantity_change: Integer
+reason: String
+notes: String
+created_by: Integer
+log_movement(): void
0..*
performs
0..*
OrderItem
+id: Integer
+order_id: Integer
+product_id: Integer
+quantity: Integer
+unit_price: Float
+total_price: Float
+getSubtotal(): Float
1..*
Role
+id: Integer
+name: String
+description: String
1
assigns
0..*
User
+id: Integer
+email: String
-password_hash: String
+full_name: String
+role_id: Integer
+is_active: Boolean
+login(): Boolean
+logout(): void
0..1
1
asks
0..*
AIQuery
+id: Integer
+user_id: Integer
+query_text: Text
+response_text: Text
+intent: String
+askAssistant(): String
1
records
0..*
Expenses
+id: Integer
+description: String
+amount: Float
+category: String
+date: Date
+notes: String
+created_by: Integer
+recordExpense(): void
Admin
+manageUsers(): void
+viewSystemReports(): void
+configureSystem(): void
Accountant
+generateInvoice(): void
+recordPayment(): void
+manageExpenses(): void
Manager
+department: String
+adjustStock(): void
+approveExpenses(): void
+viewReports(): void
1
creates
contains
0..*
Order
+id: Integer
+order_number: String
+customer_id: Integer
+created_by: Integer
+status: String
+total_amount: Float
+notes: String
+calculateTotal(): Float
+updateStatus(new_status: String): void
1
1
generates
0..1
Invoice
+id: Integer
+invoice_number: String
+order_id: Integer
+status: String
+total_amount: Float
+paid_amount: Float
+due_date: Date
+recordPayment(amount:Float) : void
+downloadPdf(): FlleBuffer
Customer
+id: Integer
+name: String
+email: String
+phone: String
+address: String
+company: String
+gstin: String
+outstanding_balance: Float
+upstandingBalance(amount: Float): void
1
places
0..*